



Ionospheric Closure of Field-Aligned Currents and its Space Weather Effects

Octav Marghita, Adrian Blăgău, Costel Bunescu, Vlad Constantinescu, Horia Comișel, Dragoș Constantinescu

Institute for Space Sciences, Bucharest, Romania

Abstract

The ionospheric closure of field-aligned currents (FACs), controlled by current continuity and Ohm's law, plays a key role in the magnetosphere-ionosphere-thermosphere coupling and, at the same time, is rich in space weather effects. The FACs' dynamics is closely tied to the variability of ionospheric currents, which drives in turn geomagnetically induced currents, potentially harmful in particular during geomagnetic storms and substorms. The FACs' closure results also in Joule heating, enhanced atmospheric density, and increased atmospheric drag, with impact on orbital motion of both spacecraft and space debris. Proper monitoring of FACs and a thorough understanding of their ionospheric closure are essential for the risk assessment of such effects and to prevent the damage of ground and space assets.

The science of FAC closure

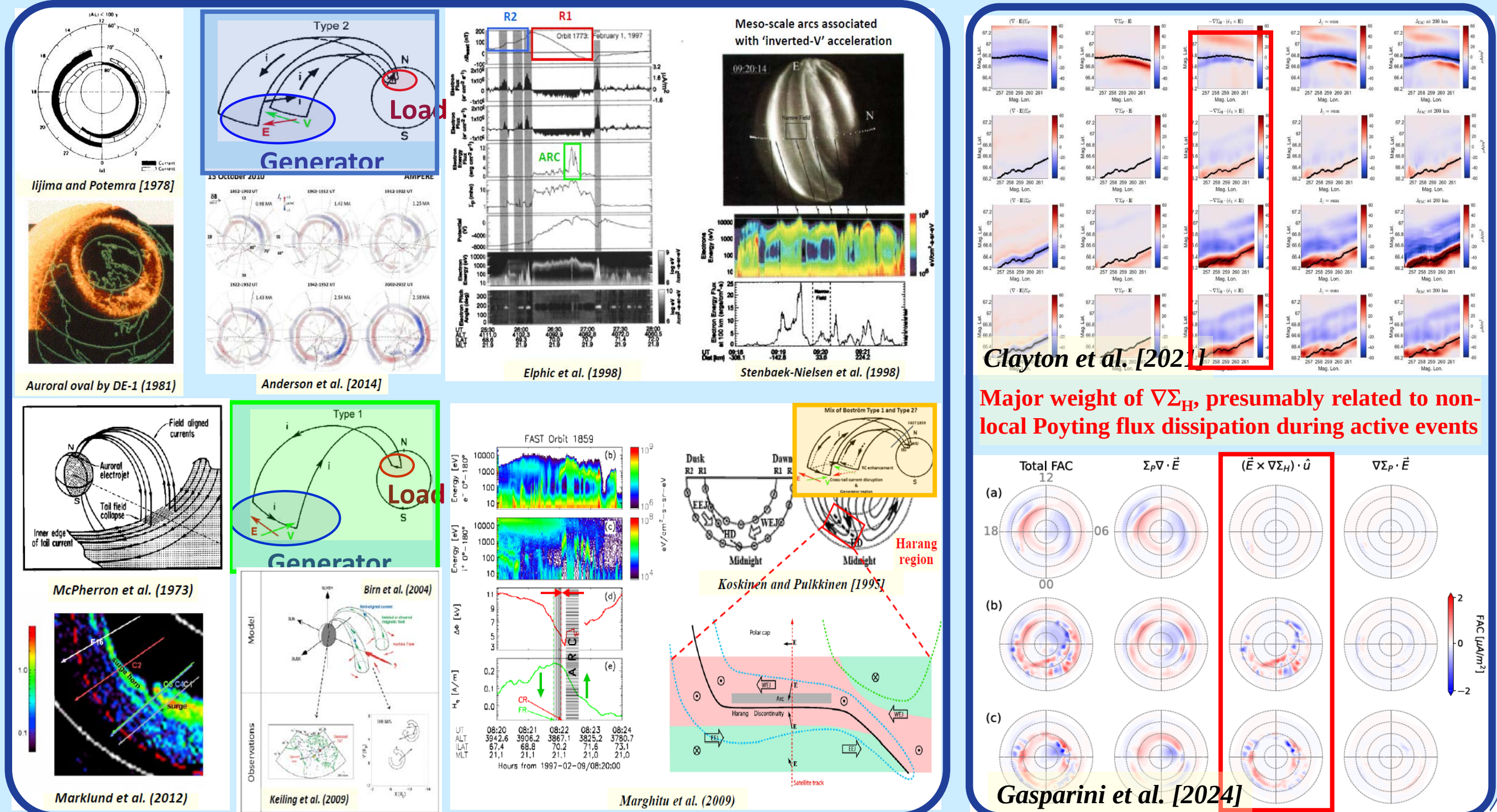
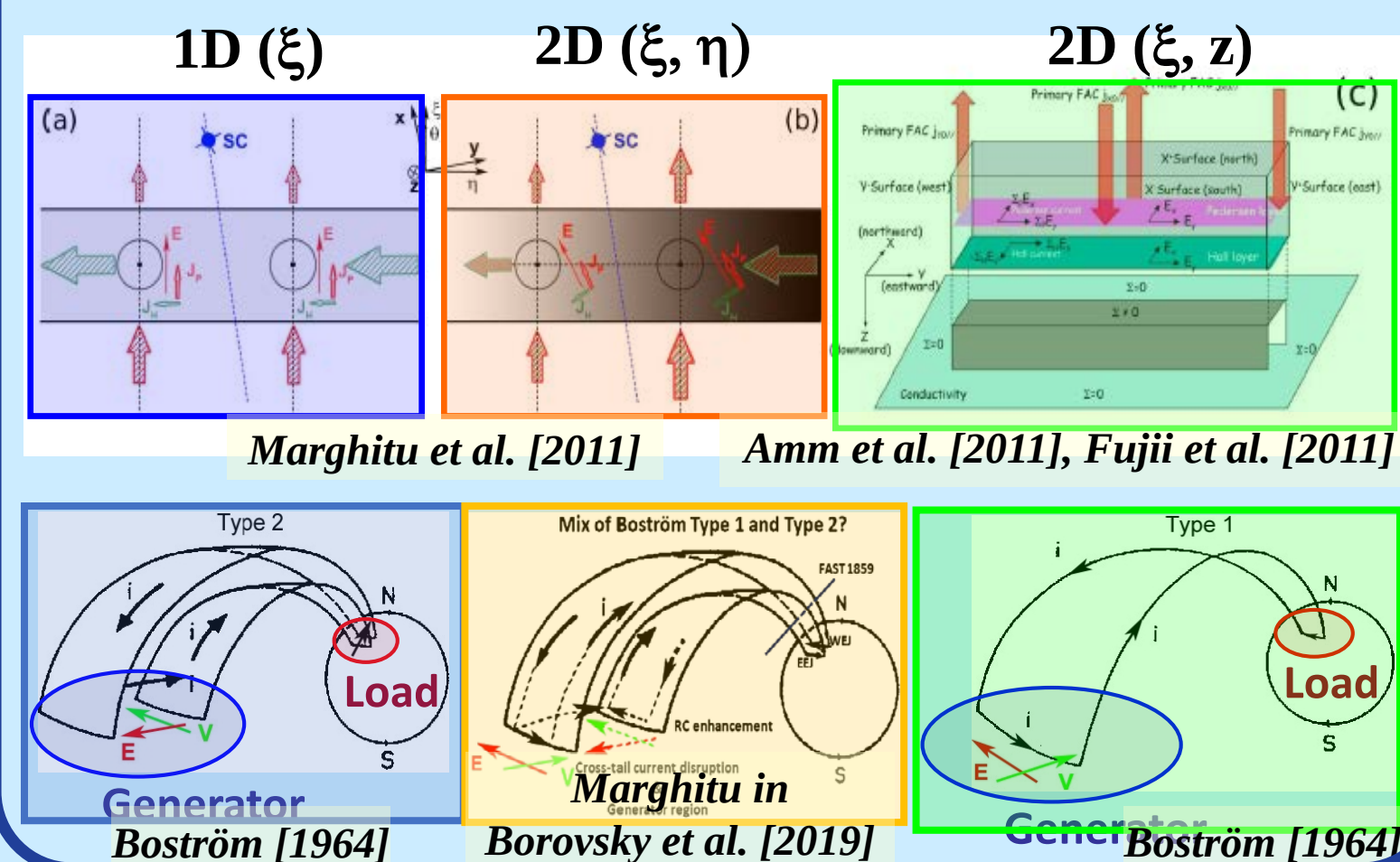
$$\mathbf{j}_{\parallel} = \Sigma_P \nabla \cdot \mathbf{E} + \nabla \Sigma_P \cdot \mathbf{E} - (\nabla \Sigma_H \times \mathbf{E})_{\parallel} - \Sigma_H (\nabla \times \mathbf{E})_{\parallel}$$

Usually neglected, however, provides inductive energy to the divergence free Hall current (Yoshikawa, 2002a, b)

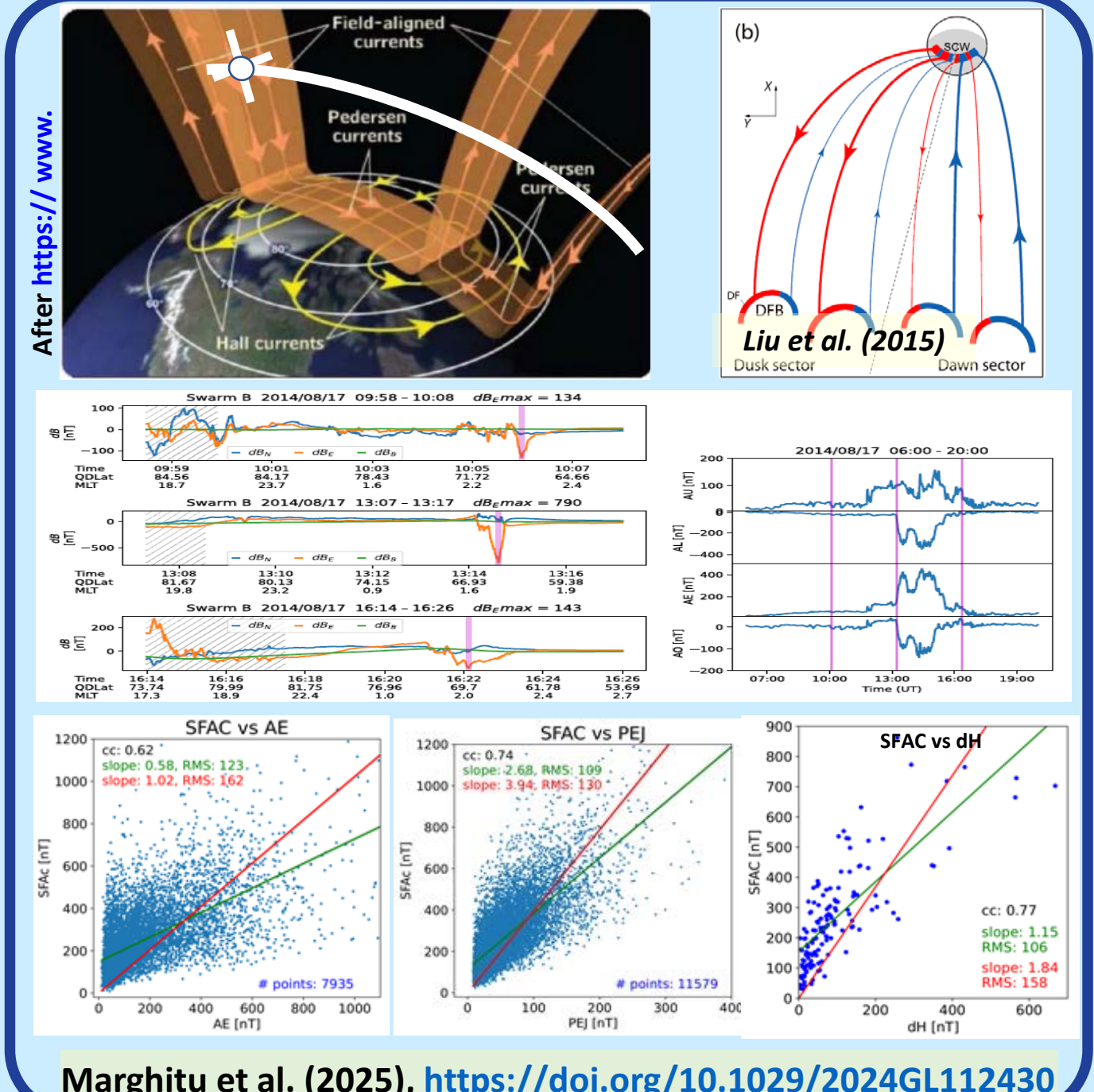
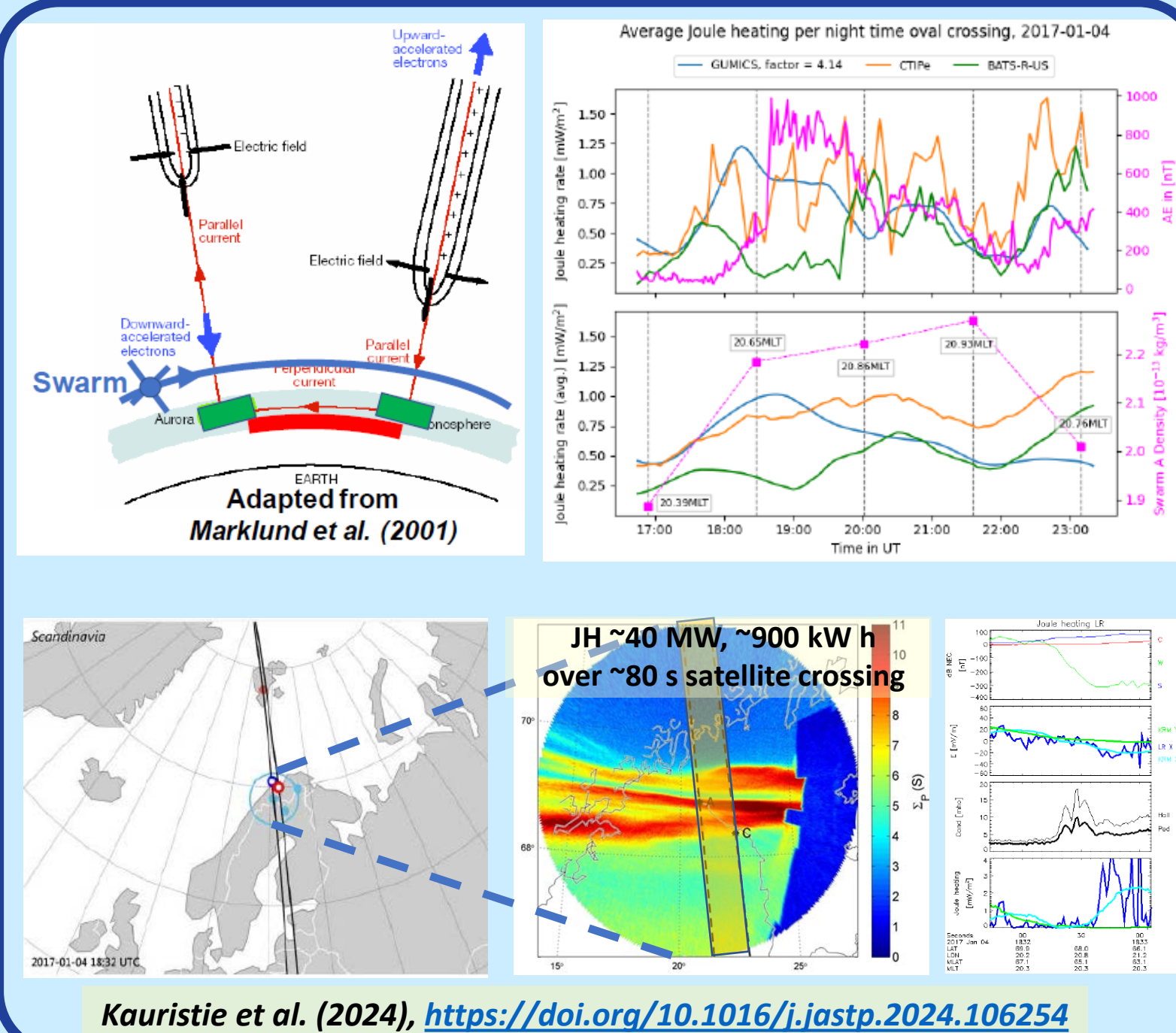
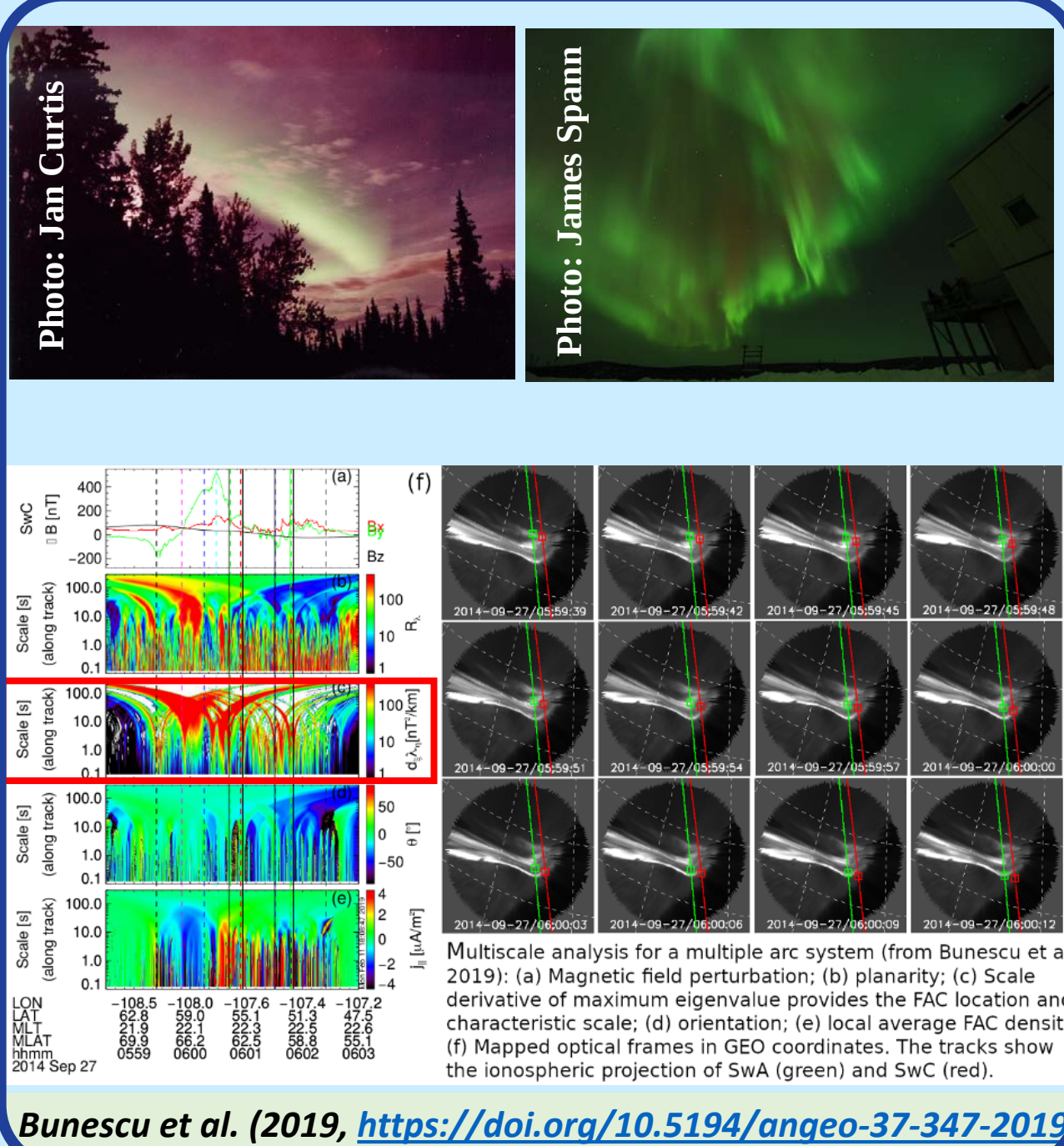
$\partial E_{\eta} / \partial \eta$ neglected

Cowling

$$\mathbf{j}_{\parallel} = \Sigma_P \nabla \cdot \mathbf{E} + \partial \Sigma_P / \partial \xi E_{\xi} + \partial \Sigma_P / \partial \eta E_{\eta} - \partial \Sigma_H / \partial \xi E_{\eta} + \partial \Sigma_H / \partial \eta E_{\xi}$$



Space weather effects



Discussion

- Sheet and filamentary (Boström Type 2 and 1) FAC configurations appear to be associated with the quiet arc/oval and active structures (BBF, SCW), respectively, consistent intuitively with prevalence of 1D/2D symmetry for quiet/active events.
- Symmetry transition from 1D to 2D appears associated with the development of longitudinal gradients. This is intuitive too, since 1D symmetry changes to 2D when the longitudinal and latitudinal length scales become comparable.
- The transition from 1D to 2D looks consistent with a mix of Boström Type 2 and Type 1 current circuits, implying asymmetric ring current enhancement and reduction / disruption of the cross-tail current, as observed.
- The Hall current appears to make a major contribution to the FAC closure during active events, on both arc and oval scales, indicating non-local dissipation of the Poynting flux. This may provide a promising investigation path for EZIE data.
- A systematic examination is still tbd, to take advantage of all available means: past missions (e.g., FAST), ongoing multi-point missions (e.g., Swarm, EZIE), multi-point rocket and/or CubeSat data, appropriate data assimilation techniques (e.g., LOMPE) and simulation codes (e.g., GEMINI).
- FAC closure drives intense Joule heating and by that enhanced thermospheric density and drag, with major impact on orbital motion of satellites and space debris. For now, models and observations are just in rough agreement, which needs swift improvement for the increasingly challenging operational needs.
- FAC is also correlated with the ionospheric current, whose variability drives ground magnetic perturbations and geomagnetically induced currents (GICs). FAC monitoring may provide an efficient mean to quantify the risk of harmful GICs.

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Ionospheric Closure of Field-Aligned Currents: Mechanisms and Impacts in Geospace and Beyond

This Research Topic will advance our understanding of the multidimensional and dynamic aspects of ionospheric closure of FACs. Primary objectives include identifying how different closure pathways operate under various conditions, clarifying the roles of associated currents and atmospheric coupling, and improving models that predict energy transfer and space weather impacts. By combining empirical observations, simulations, and theoretical approaches, this Research Topic seeks to bridge gaps in current knowledge and inform both practical applications and fundamental research across geospace and planetary environments.

The scope includes studies focused on FAC closure mechanisms within Earth's I-T system and in other planetary contexts. We invite contributions addressing core theoretical principles, advanced modelling, and observational insights, provided they relate to the three dimensional FAC closure.

Topic editors

Dr Octav Marghita
Space Science Institute
Romania

Dr Jesper W Gjerloev
The Catholic University of
America
United States

Prof Tomas Karlsson
Royal Institute of Technology
Sweden

Prof Kristina A Lynch
Dartmouth College
United States



Submission deadlines

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For more information

spacesciences@frontiersin.org
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